1. Title of the Project / Product

FrostChain E-Skin Crate: A Tamper-Aware, AI-Verified Cold-Chain Crate for the First Mile

FrostChain Smart Crate (PCM + solar Peltier cooling, 5G/blockchain quality scoring) fused with Graphite E-Skin (physics-compensated strain & tamper sensing, FPGA ratiometric Σ-Δ ADC, INT8 edge AI).

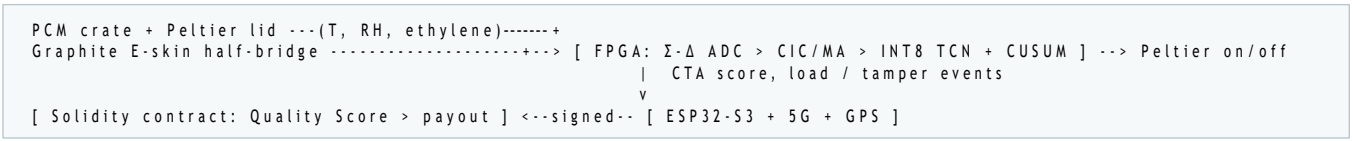
2. Problem Statement

> **Roughly 30% of India's fruit and vegetable output is lost in the first-mile cold-chain void because no low-cost, off-grid crate both holds produce at ~4 °C and produces tamper-proof evidence of its temperature and handling history that can settle payment fairly.**

3. Solution Proposed

One crate, four layers – each hardware layer feeds the next, and the AI sits where it turns cheap parts into trustworthy data:

- **Cool (FrostChain):** EPS/VIP crate lined with PCM packs (paraffin or salt hydrate, 4–6 °C melt) acts as a passive "thermal battery". A top-mounted solar Peltier lid (TEC1-12706, 20–40 W flexible panel, LiFePO4) re-stabilises the interior only when it drifts to 5 °C. Fully off-grid.
- **Sense (Graphite E-Skin):** a printed graphite-tape *active/dummy half-bridge* on the lid seal and walls senses strain, rough handling, lid opening and tamper at near-zero cost. The half-bridge cancels common-mode thermal/humidity drift (≈ 20× at 5% mismatch). A **ratiometric Σ-Δ ADC in Verilog** (FPGA comparator input, 1-bit feedback GPIO, CIC³ + moving-average decimation with a 50 Hz null) digitises it – and the same supply-independent ADC also reads the crate's thermistor and humidity channels.
- **Decide (edge AI):** a fixed-point INT8 TCN/MLP on the FPGA (≈ 1.7 k parameters, ≈ 25 kMAC) removes residual graphite drift and creep, separates *slow thermal drift and reversible load* from *irreversible steps* (crack, peel, tamper), and predicts PCM hold-time so the Peltier fires early only when needed. A CUSUM guard + consensus gate suppresses false alarms.
- **Settle (5G + blockchain):** an ESP32-S3/5G module with GPS pushes signed event records to a Solidity smart contract. Quality Score = 100 − $\Sigma(T_{actual} - T_{target})\cdot t$ − handling penalty; a tamper flag places the payout under review. Payout: score > 90 → 100%; 70–90 → 80%; ≤ 70 → 50% base.



4. Implementation Plan

Phase	Work	Exit criterion
1. Modelling & SPICE (Wk 1–4)	SPICE behavioural model of graphite gauge (drift, creep, damage step); half-bridge + Σ-Δ loop with Monte-Carlo; lumped thermal-RC model of crate + PCM + Peltier to size PCM mass and solar panel; synthetic dataset.	Simulated drift suppression ≥ 15×; ADC ≥ 15 ENOB at 10 Hz (sim); crate holds 4 ± 2 °C ≥ 8 h at 40 °C ambient (sim).
2. Verilog & hardware (Wk 5–9)	Verilog Σ-Δ front end, CIC³, MA-5, phase/chop sequencer; bit-exact Python golden model; FPGA synthesis and timing closure. Build E-skin tape pairs and the PCM/Peltier crate.	≥ 13 ENOB measured (resistor-emulated bridge); RTL = golden model; crate prototype cools under load.
3. Edge AI & integration (Wk 10–14)	Train TCN/MLP; INT8 quantisation-aware training; Verilog inference engine (one time-multiplexed MAC); CUSUM guard; ESP32-S3 uplink; Solidity Quality-Score contract on a testnet.	INT8 within 1% of FP32; zero mismatch vs golden vectors; ≤ 20 με RMS after correction; latency ≤ 2 s.
4. Validation & demo (Wk 15–18)	Heat/humidity cycling, load and peel/cut tamper tests, unseen tape lots; end-to-end run: crate → tamper event → on-chain payout change.	False alarms ≤ 1 per 1000 device-hours; tamper step ≥ 100 με-equivalent detected; live demo.

- **Beneficiaries:** smallholder farmers and FPOs, first-mile aggregators, buyers/retailers, and pharma or berry/dairy shippers who need audited cold-chain proof.

- **Outcomes:** lower spoilage, fair quality-linked pricing, fewer disputes, tamper-evident audit trail from farm gate to collection centre.
- **Deliverables:** working crate prototype, Verilog RTL + golden model, SPICE decks, quantised model, smart contract, test report, demo.

5. Safety and Efficiency Analysis

- **Produce safety:** 4 °C setpoint held passively by PCM; Peltier is the backup, so temperature excursions are corrected before they accumulate as thermal abuse.
- **Asset integrity:** E-skin flags lid opening, crushing, drops and tamper; drift/creep are modelled so they are never mistaken for damage. A tamper alarm requires network posterior + CUSUM persistence + healthy dummy trace; a separate FAULT state covers broken sensors.
- **Electrical safety:** low-voltage (12 V) LiFePO4 pack with BMS over-current/over-temperature cut-off; Peltier duty-limited; saturating fixed-point arithmetic, watchdog and CRC on records.
- **Efficiency:** PCM does the bulk cooling, so solar energy is spent only on top-up. Bridge is excited only during ≈ 20 ms conversions (≈ 2% duty, ≈ 5 µW average); one inference ≈ 2 ms at 12 MHz; only events and the Quality Score are transmitted.
- **Data trust:** signed, hash-anchored records mean neither farmer nor buyer can alter the temperature or handling history.

6. Scalability and Future Development

- **Manufacturing:** E-skin printed on labels or corrugated board in one pass (better pair matching); retrofit lid module for existing crates; PCM melt point tuned per cargo (2–8 °C medicines, ≈ 4 °C berries, 10–13 °C other produce).
- **Silicon:** FPGA → custom mixed-signal ASIC (ratiometric Σ-Δ + INT8 MAC, ≤ 10 µW sensing) reusing the same Verilog; multi-channel nodes for pallets and reefers.
- **Ecosystem:** fleet-level models per tape lot and crop; vaccine, dairy and pharma cold chain; insurance and carbon-credit claims from verified spoilage avoided.

7. Innovation Advantage

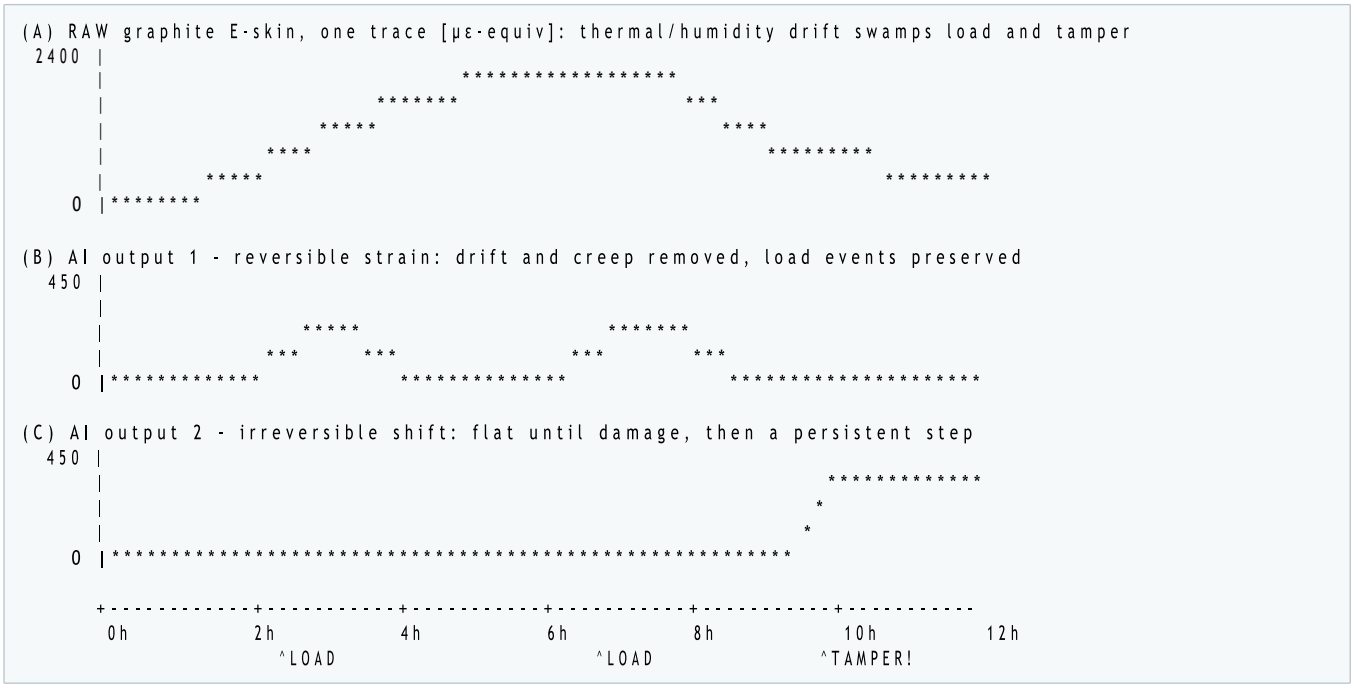
	Insulated box + ice	Data-logger / reefer	FrostChain E-Skin Crate
Off-grid active cooling	No (melts)	Grid/diesel	PCM + solar Peltier
Handling & tamper evidence	None	Temperature only	Strain, shock, tamper - classified
Sensing cost	–	High (logger + IMU)	Near-zero printed graphite + AI
Payment-linked proof	No	Manual claims	Smart-contract Quality Score

- **AI fixes bad physics:** graphite tape is unusable raw (percolation-network drift with temperature, humidity and creep); half-bridge plus a tiny INT8 network makes it trustworthy, so integrity sensing costs almost nothing.
- **Own the silicon:** the ADC is a Verilog design, not a part – ratiometric, reference-free, and reused for every sensor in the crate and later in an ASIC.
- **Evidence, not just temperature:** the Quality Score combines thermal history with verified handling, which is what makes automated fair pricing credible.

8. Visual Impact & User Journey

Stage	BEFORE	AFTER - FrostChain E-Skin Crate
Harvest & load	Berries in open crates in the sun; no cooling before the collection centre.	Loaded into a pre-cooled PCM crate; E-skin baselines in seconds.
Transit	Heat and rough handling silently ruin produce; no record.	PCM holds ~4 °C; Peltier tops up on solar; drift ignored, drops and load logged.
Tampering	Crate opened or swapped; dispute has no evidence.	Irreversible step detected; TAMPER event signed on-chain; payout held.
Settlement	Buyer haggles by eye; farmer is underpaid.	Contract pays by verified Quality Score; farmer sees why.

How the AI separates thermal drift from a tamper event



Illustrative schematic from the behavioural model (5% mismatch, creep, two load events, tamper step at ≈ 9.7 h); to be replaced with measured traces in Phase 4.